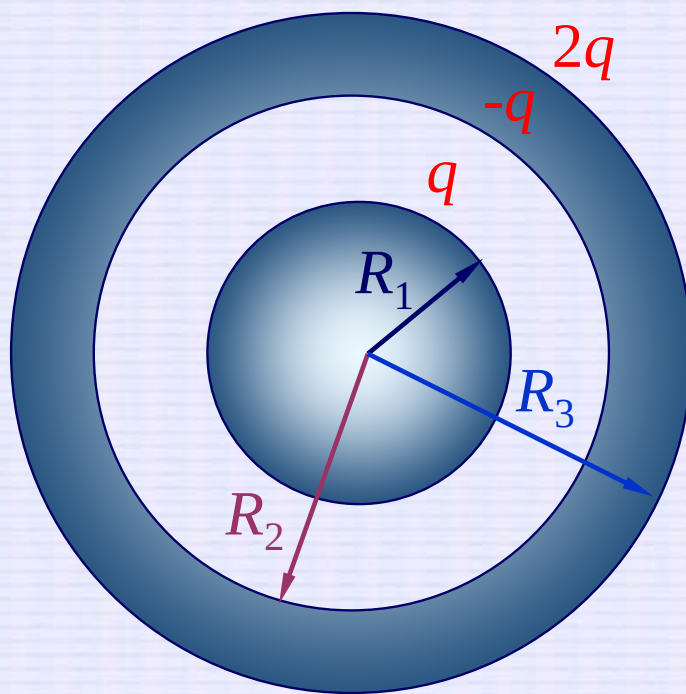


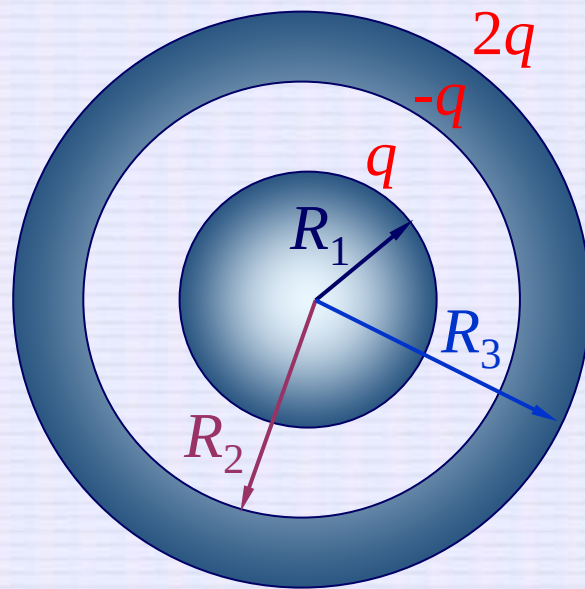
例 8-1 一半径为 R_1 的金属球置于内、外半径分别为 R_2 和 R_3 的金属球壳中心，金属球和球壳均带有电量为 q 的正电荷。求：

- (1) 金属球和球壳的电势；
- (2) 若把外球壳接地，则内球和外球壳的电势分别为多少？
- (3) 若把内球接地，则金属球和球壳的电势分别为多少？



解：

$$(1) \quad E = \begin{cases} 0 & (r < R_3) \\ \frac{q}{4\pi\epsilon_0 r^2} & (R_3 < r < R_2) \\ 0 & (R_2 < r < R_1) \\ \frac{2q}{4\pi\epsilon_0 r^2} & (r > R_1) \end{cases}$$

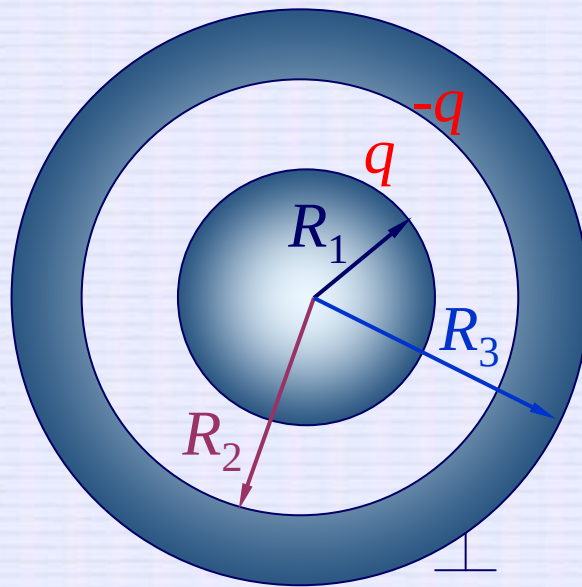


$$\begin{aligned} V_1 &= \int_{R_1}^{\infty} \mathbf{E} \cdot d\mathbf{l} = \int_{R_1}^{\infty} E(r) dr = \int_{R_1}^{R_2} \frac{q}{4\pi\epsilon_0 r^2} dr + \int_{R_3}^{\infty} \frac{2q}{4\pi\epsilon_0 r^2} dr \\ &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} + \frac{2}{R_3} \right) \end{aligned}$$

$$V_2 = \int_{R_3}^{\infty} E(r) dr = \int_{R_3}^{\infty} \frac{2q}{4\pi\epsilon_0 r^2} dr = \frac{q}{2\pi\epsilon_0 R_3}$$

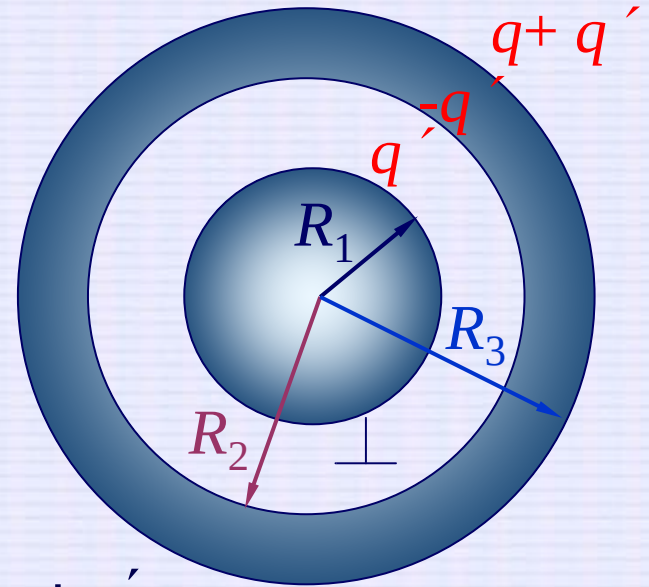
$$(2) V_2 = 0$$

$$V_1 = \int_{R_1}^{R_2} \frac{q}{4\pi\epsilon_0 r^2} dr = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$



(3)

$$E = E(r) = \begin{cases} 0 & (r < R_1) \\ \frac{q'}{4\pi\epsilon_0 r^2} & (R_1 < r < R_2) \\ 0 & (R_2 < r < R_3) \\ \frac{q + q'}{4\pi\epsilon_0 r^2} & (r > R_3) \end{cases}$$



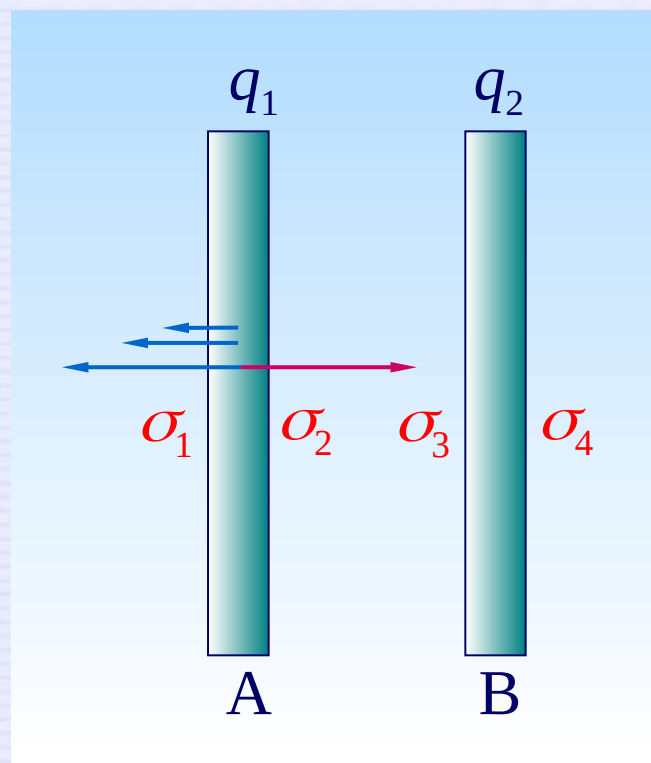
$$V_1 = \int_{R_1}^{\infty} E(r) dr = \int_{R_1}^{R_2} \frac{q'}{4\pi\epsilon_0 r^2} dr + \int_{R_3}^{\infty} \frac{q + q'}{4\pi\epsilon_0 r^2} dr$$

$$= \frac{1}{4\pi\epsilon_0} \left(\frac{q'}{R_1} - \frac{q'}{R_2} + \frac{q + q'}{R_3} \right) = 0$$

$$q' = - \frac{1/R_3}{1/R_1 - 1/R_2 + 1/R_3} q$$

$$V_2 = \int_{R_3}^{\infty} \frac{q + q'}{4\pi\epsilon_0 r^2} dr = \frac{q + q'}{4\pi\epsilon_0 R_3} = \frac{q}{4\pi\epsilon_0 R_3} \frac{1/R_1 - 1/R_2}{1/R_1 - 1/R_2 + 1/R_3} q$$

例 8-2 两块大导体平板，面积为 S ，分别带电 q_1 和 q_2 ，两板间距远小于板的线度。求平板各表面的电荷密度。



解：电荷守恒

$$\sigma_1 S + \sigma_2 S = q_1$$

$$\sigma_3 S + \sigma_4 S = q_2$$

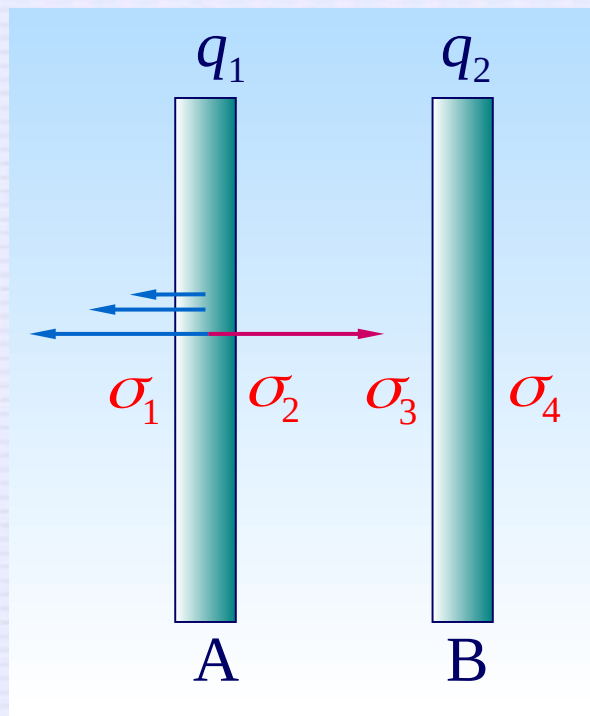
导体板内 $E = 0$

$$E_A = \frac{\sigma_1}{2\epsilon_0} - \frac{\sigma_2}{2\epsilon_0} - \frac{\sigma_3}{2\epsilon_0} - \frac{\sigma_4}{2\epsilon_0} = 0$$

$$E_B = \frac{\sigma_1}{2\epsilon_0} + \frac{\sigma_2}{2\epsilon_0} + \frac{\sigma_3}{2\epsilon_0} - \frac{\sigma_4}{2\epsilon_0} = 0$$

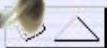
$$\sigma_2 = -\sigma_3 = \frac{q_1 - q_2}{2S}$$

$$\sigma_1 = \sigma_4 = \frac{q_1 + q_2}{2S}$$



作业：

6-2 、 6-7



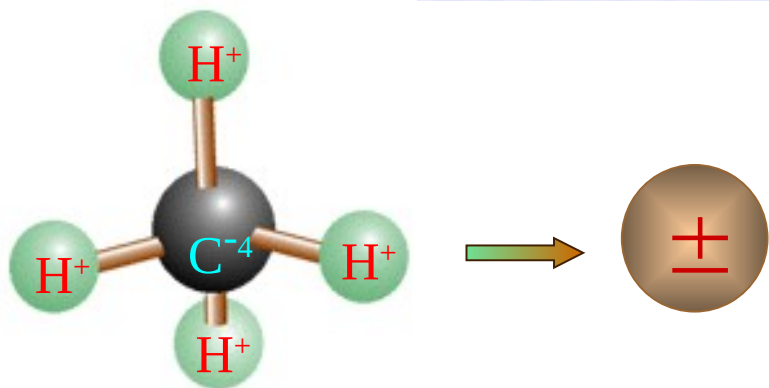
§8-2 静电场中的电介质

1. 电介质的极化

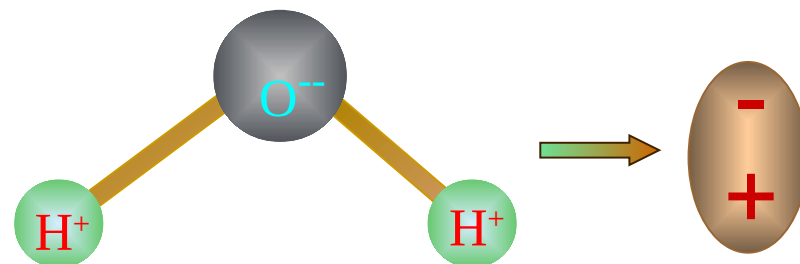
1.1 电介质

分子中的正负电荷束缚很紧，介质内部几乎没有自由电荷。

电介质 { 无极分子电介质
有极分子电介质

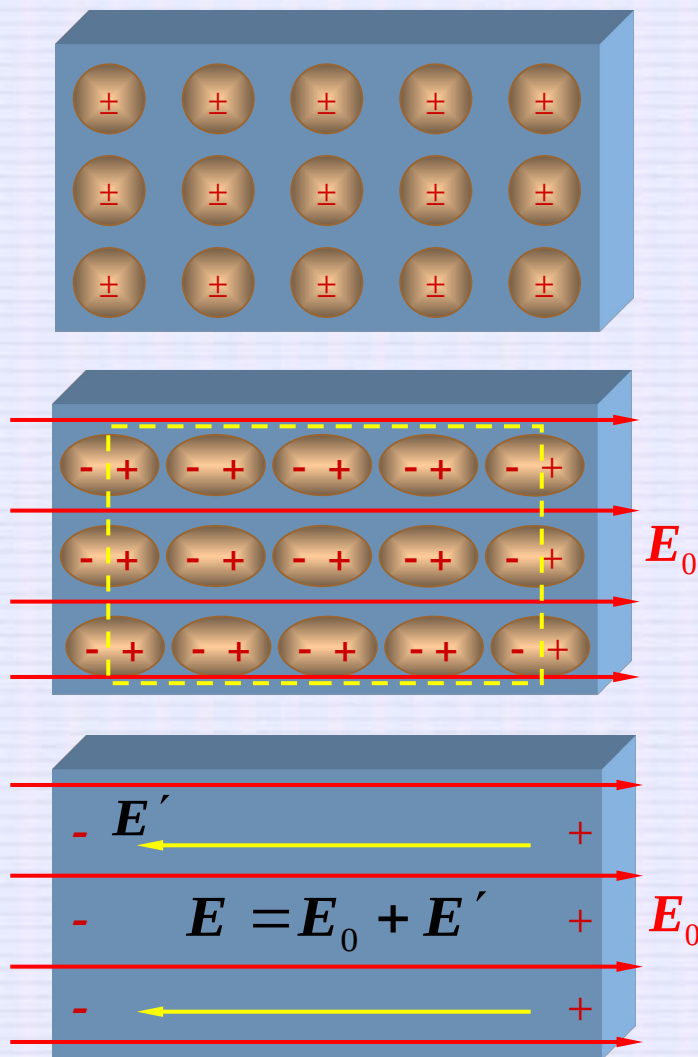


无极分子：CH₄

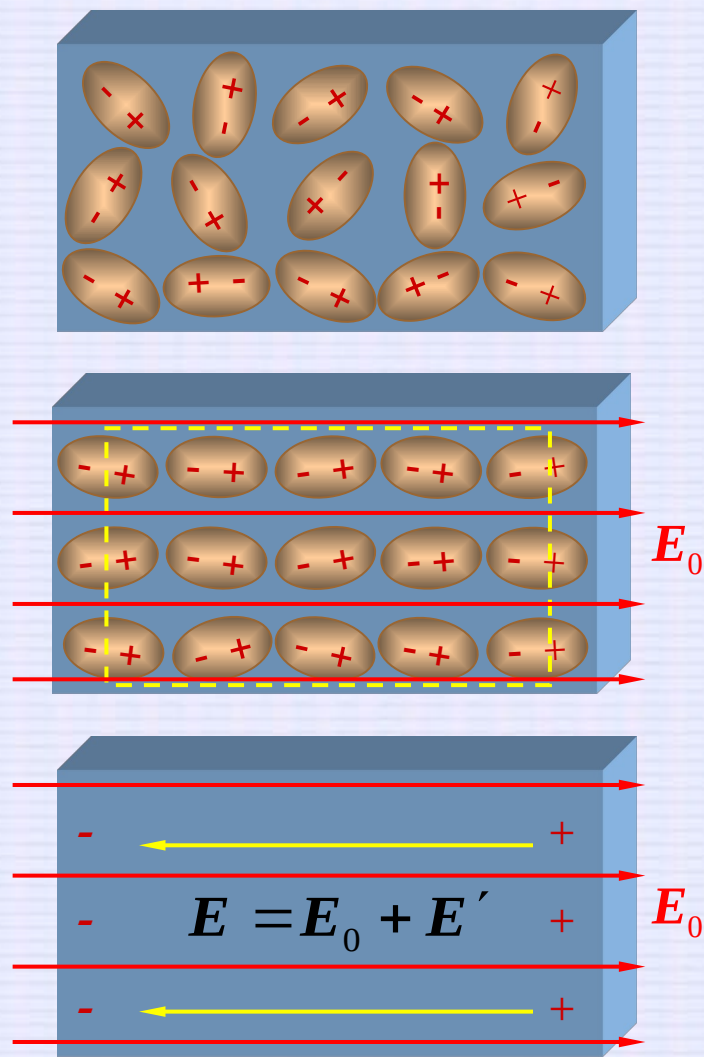


有极分子：H₂O

1.2 电介质的极化过程

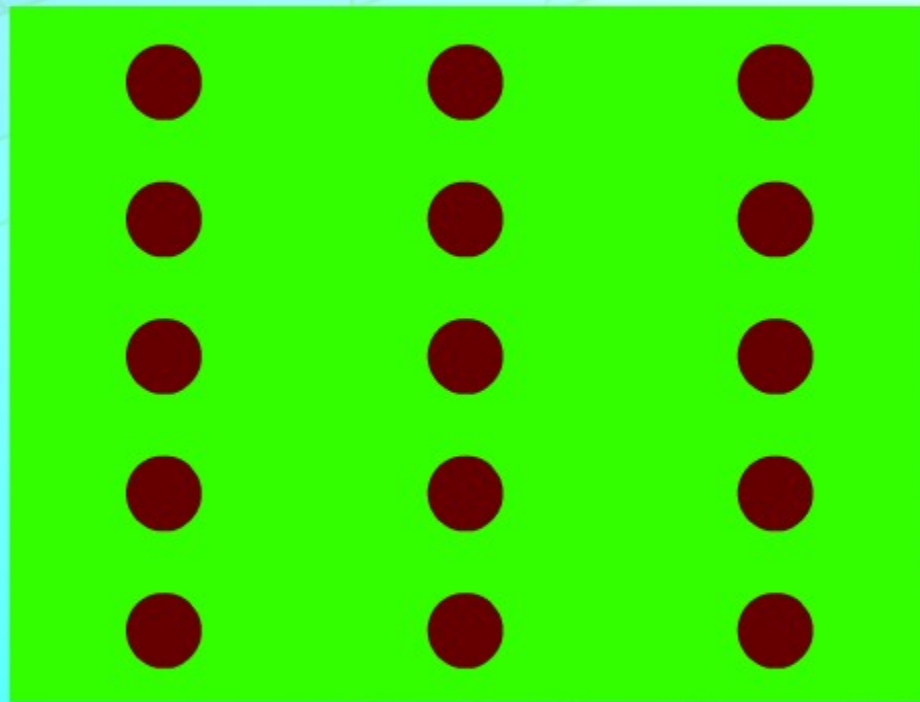


无极分子的位移极化

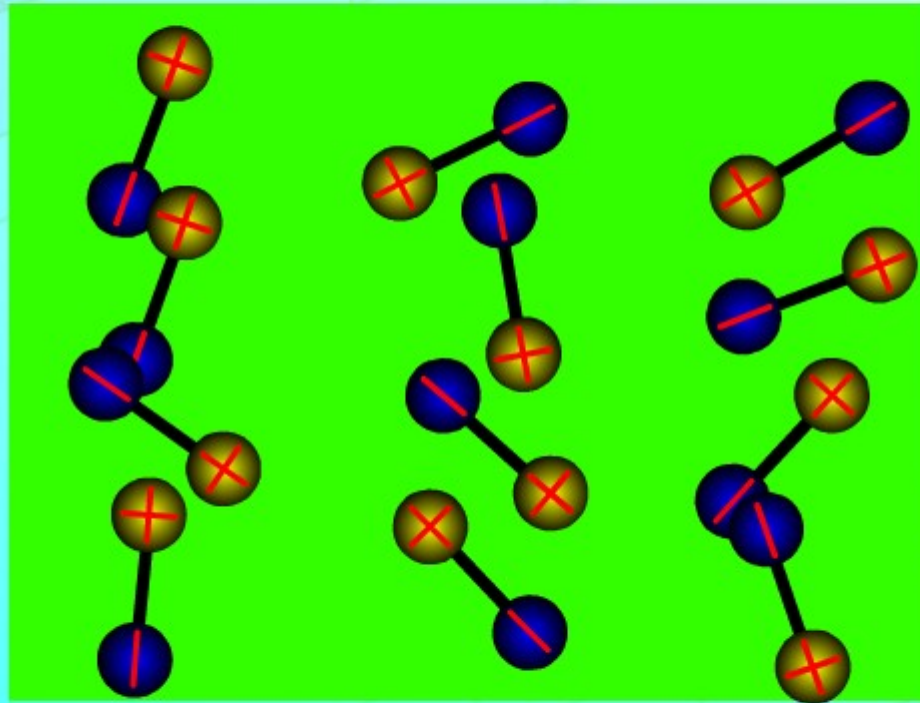


有极分子的转向极化

位移极化



转向极化



1

无外电场



➤ **电介质的极化：**

在外电场的作用下，介质表面产生电荷的现象称为电介质的极化。

➤ **电介质的极化的结果：**

$$\left\{ \begin{array}{l} \text{产生极化电荷 } q' \\ \text{极化电荷产生电场 } E' \rightarrow E = E_0 + E' \end{array} \right.$$

2. 极化强度与极化电荷

2.1 极化强度

描述电介质的极化程度的物理量。

$$\vec{P} = \frac{\sum_i p_i}{\Delta V} \quad (\text{Cm}^{-2})$$

设 ΔV 内各分子电偶极矩都相同： $p_i = p$ ， 则

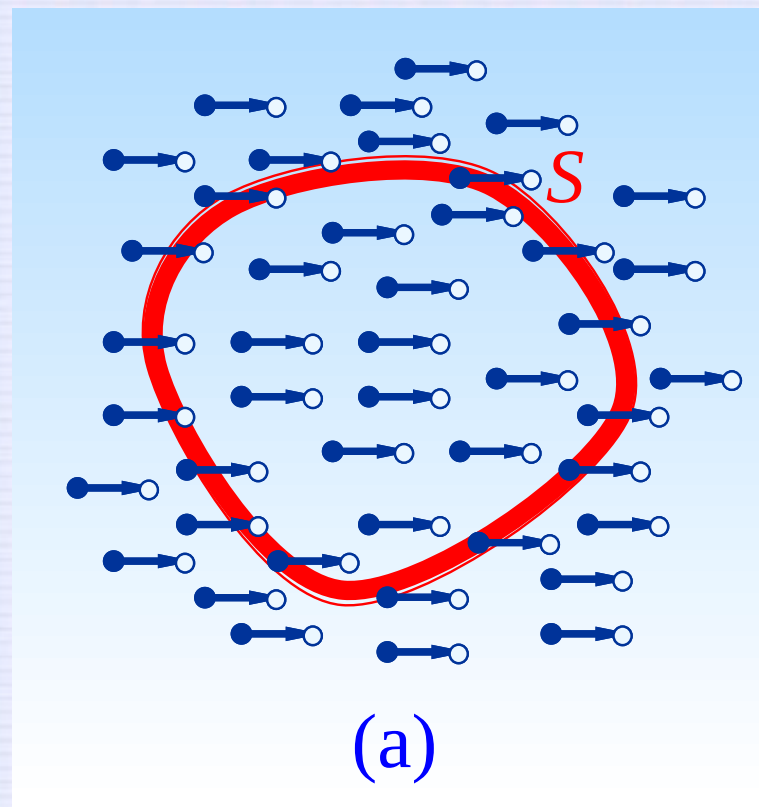
$$P = np$$

➤ 电极化强度为矢量。

2.2 极化电荷与极化强度的关系

极化体电荷与极化强度的关系：

- 只有跨过闭合面边界的分子对闭合面内极化电荷代数和有贡献
- 闭合面内极化电荷代数和等于因极化而穿过闭合面边界的电荷代数和（“穿出”的贡献为正，“穿入”的贡献为负）的负值。



因极化而穿过 dS 的电荷：

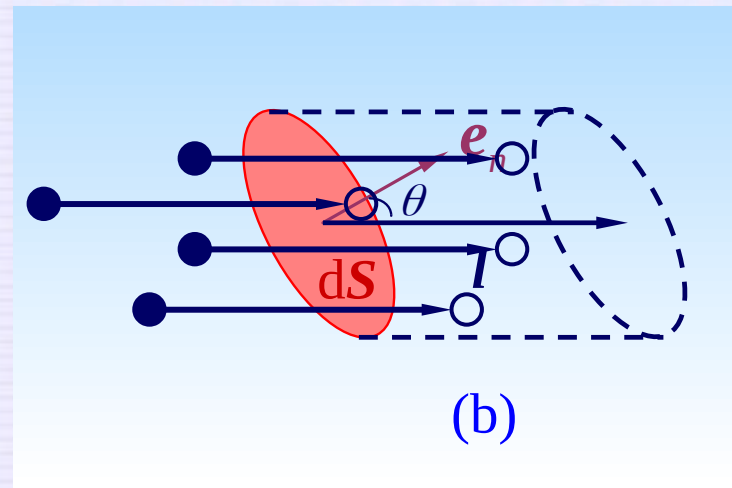
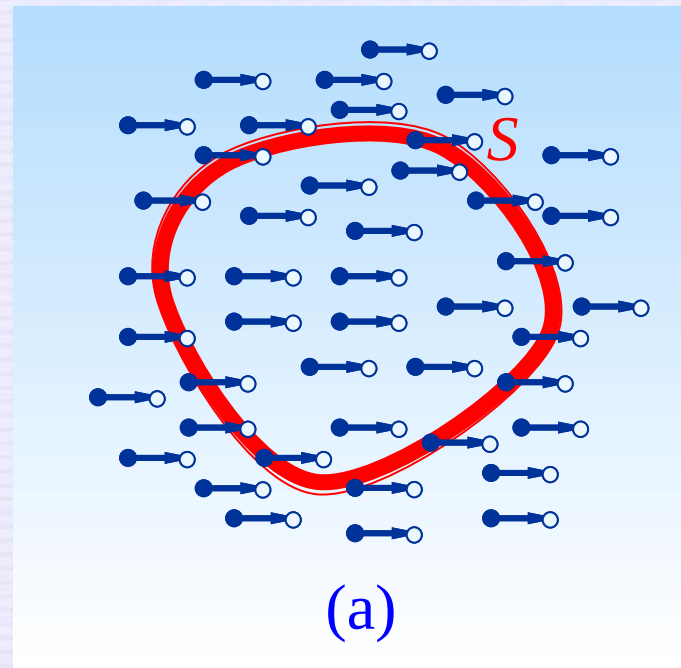
$$nql \cos \theta dS = np \cos \theta dS = P \cdot dS$$

因极化而穿过闭合面 S 的电荷：

$$\oiint_S P \cdot dS$$

闭合面 S 内的极化电荷 $\sum_i q'_i$
满足：

$$\oiint_S P \cdot dS = - \sum_i q'_i$$



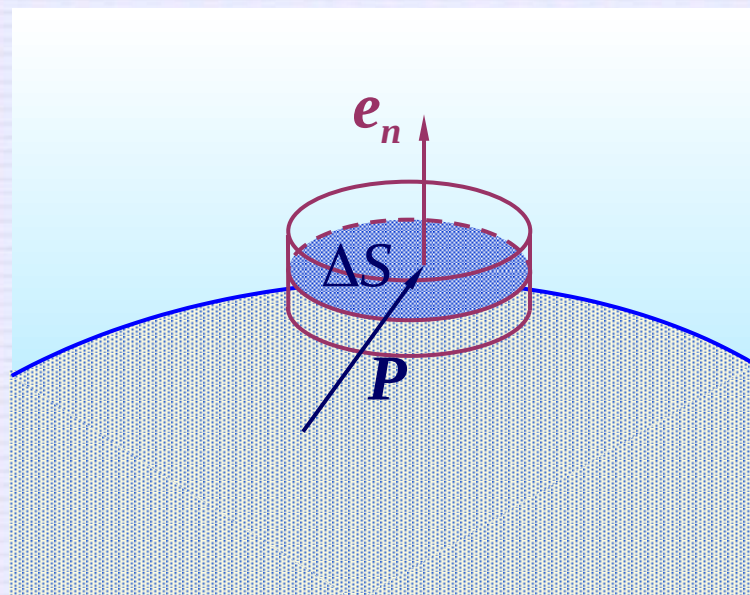
极化面电荷与极化强度的关系：

$$\oint_S \vec{P} \cdot d\vec{S} = \vec{P} \cdot \Delta\vec{S} (-\vec{e}_n) \\ = -\vec{P} \cdot \vec{e}_n \Delta S$$

$$\sum_i q'_i = \sigma' \Delta S$$

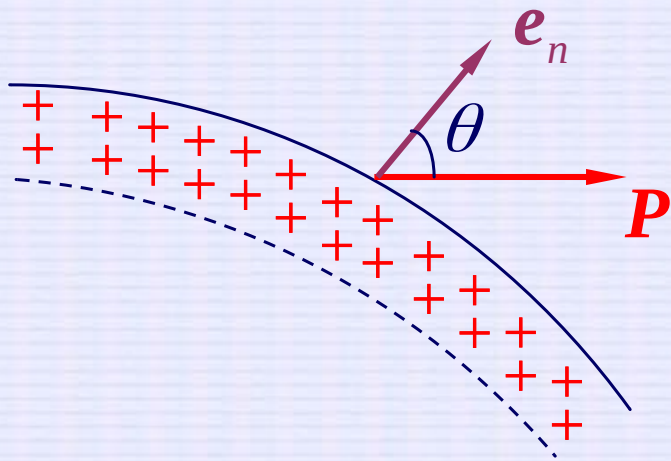
$$-\vec{P} \cdot \vec{e}_n \Delta S = -\sigma' \Delta S$$

$$\sigma' = \vec{P} \cdot \vec{e}_n = P_n = P \cos \theta$$



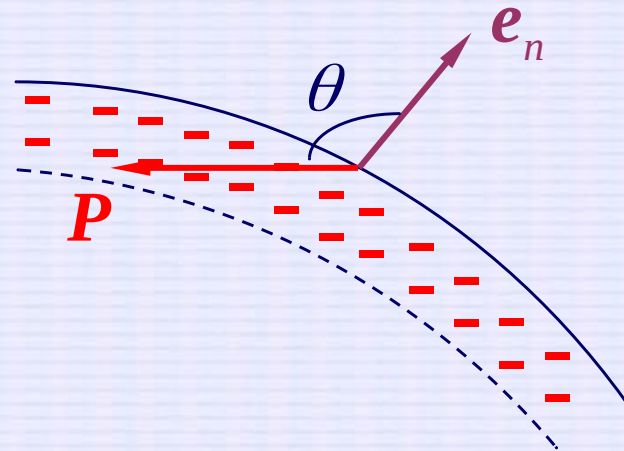
$$\sigma' = P \cos \theta = P \cdot e_n$$

$$\sigma' = P \cos \theta > 0$$



(a)

$$\sigma' = P \cos \theta < 0$$



(b)

例 8-3 半径为 R 的介质球被均匀极化，极化强度为 P 。求：

- (1) 电介质球表面极化电荷的分布；
- (2) 极化电荷在球心处所激发的场强。

